Learned Recurrence and Temporal Memory in Event-Based Spiking Networks

Deep recurrent SNNs vs. LSMs on event-based gesture sequences

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Problem: Spiking networks encode time but don't retain it

1. **Recognizing a gesture requires watching it over time**

   An event camera captures changes as your hand moves, but recognizing the gesture requires information over time.

2. **Most deep SNNs have short-lived memory**

   Their only memory is membrane leak, which fades quickly once input stops.

3. **Fixed connections limit learned memory**

   In LSMs, internal connections are fixed and only the output layer is trained. This lets the network learn to interpret its activity, but not adjust what it remembers.

Motivation

Low-power edge systems (robots, drones, and brain–machine interfaces) need to integrate evidence over time.

Pfeiffer & Pfeil identify event-based sensing as a major opportunity for SNNs. They propose deep recurrent SNNs as a way to better store and integrate temporal information.

https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2018.00774/full#s2

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Question and related work

Question: On event-based gesture recognition, does a deep SNN with learned recurrent connections integrate information across time better than an LSM, and better than a feed-forward SNN relying on neuron memory alone?

Prior work	What it shows	Limitation for our question
Pfeiffer & Pfeil (2018)	Reviews deep SNN training; flags deep recurrent SNNs as a future direction	Recurrence discussed only as outlook
Maass et al. (2002); George et al. (2020)	LSMs: random spiking reservoir + trained readout, including on event data	Recurrent weights never learn
Amir et al. (2017); Fang et al. (2021)	DVS128 Gesture benchmark; strong deep feed-forward SNNs	Memory comes only from neuron dynamics
Bellec et al. (2018), LSNN	Learned recurrence in SNNs trained with BPTT	Evaluated mainly on non-event benchmarks

Gap targeted: a size-matched, same-pipeline comparison of learned vs. random vs. no recurrence, with a direct test of memory across event-free gaps.

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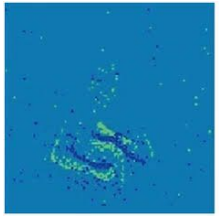
DVS128 Gesture Dataset: event-based sequences

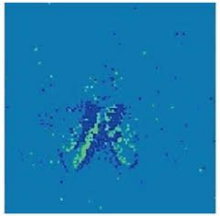
Gesture dataset

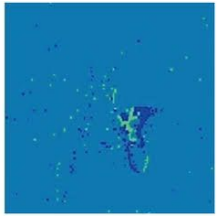
IBM DVS128 Gesture

Amir et al., CVPR 2017

- 128 × 128 event camera
- 29 people, 3 lighting conditions
- 11 gesture classes
- 1,340 gesture clips, ~6 seconds each
- Tested on unseen people
- SpikingJelly loading and preprocessing







1 Event definition

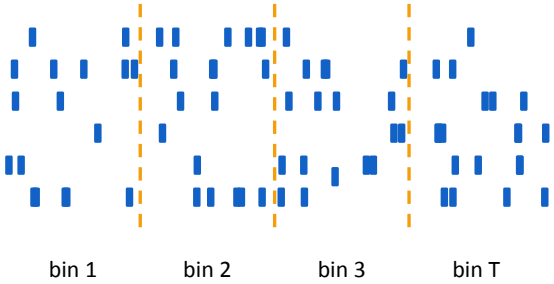
`(x, y, time, ON/OFF)`

Each event contains:

- (x,y) - which pixel changed
- time - when it changed
- ON/OFF - pixel became brighter or darker

Event camera records when individual pixels detect a change in brightness.

2 Event data as SNN input



- Slice each clip into T time bins
- Count ON/OFF events per pixel

Input tensor: `T × 2 × 128 × 128`

- SNN processes one bin per time step
- Order and timing are preserved

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Approach

DVS128 Gesture
11 classes, event streams

→

Bin events into T frames
(SpikingJelly loader)

→

Same neuron model (LIF/PLIF), same input, matched neuron counts

CONTROL

Feed-forward SNN

SpikingJelly conv + LIF stack, reproduced from the DVS Gesture tutorial

BASELINE

Liquid state machine

Large, sparse network of spiking neurons with fixed random recurrent weights; train the readout only.

OURS

Deep recurrent SNN

Same conv stack + trainable recurrent weights at 1–2 layers: spikes at t feedback at t+1

Training: surrogate-gradient BPTT (SpikingJelly + PyTorch), same epochs and optimizer across arms
Hardware: single GPU (Duke Compute Cluster or Colab)
Fairness: same model capacity across arms ⟶ isolates architecture as the variable.

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Evaluation: accuracy, memory, and spike cost

1. **Top-1 accuracy, full clips**

   Standard DVS128 Gesture test split

2. **Accuracy with inserted gaps**

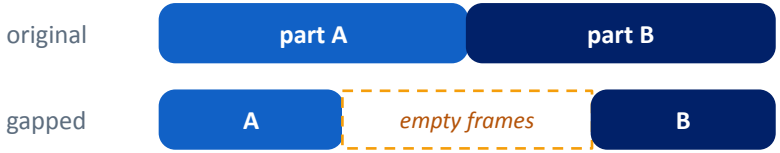
   Sweep gap length and position; no retraining

3. **Spikes per clip**

   Efficiency proxy for neuromorphic hardware

Baselines: LSM (main) and feed-forward SNN (control)

Gap-insertion test



Insert empty frames rather than erase events, so every event is still there; only the delay changes.

Definition of Success

- Feed-forward control lands near the Classify DVS Gesture tutorial's reported accuracy (pipeline check)
- Recurrent SNN beats the LSM on full clips at matched size
- Recurrent SNN loses less accuracy than both baselines as gaps grow

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Timeline, Owners, Risks

- **Sep 28** — Kickoff presentation
- **Oct 7** — Data pipeline running, feed-forward control reproduced
- **Oct 14** — LSM and recurrent SNN training end to end
- **Oct 21** — **Milestone:** all three arms trained, first full-clip comparison
- **Nov 11** — Gap-insertion sweep and spike counts
- **Nov 25** — Ablations (where recurrence goes), write-up
- **Dec 2** — **Final presentation and report**

Risk	Backup plan
Recurrent training unstable or out of GPU memory	Fewer time steps, truncated BPTT, recurrence at one layer only
LSM too weak to be a fair baseline	Tune reservoir size, sparsity, and weight scale before comparing
Compute and time limits (dataset is several GB)	Cache binned frames, downsample input, shorter training
No difference under gaps	Vary gap length and position; report and explain as a finding

Owners

Bennett: data pipeline, feed-forward control, gap eval
Carpenter: LSM baseline, spike metrics
Hao: recurrent SNN, training runs

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References

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Generative AI

Generative AI was used to format slides.

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